

CARMACKS, YT, Canada

WMO: 710390

Lat: 62.1150N

Lon: 136.1919W

Elev: 1781

StdP: 13.77

Time Zone: -8.00 (NAP)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.1	(c) -40.1	(d) -51.8	(e) 0.3	(f) -43.4	(g) -46.3	(h) 0.4	(i) -37.9	(j) 18.4	(k) 32.3	(l) 15.9	(m) 30.0	(n) 0.3	(o) 130	(p) 0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.1	80.6	57.2	77.1	55.9	73.8	55.1	60.2	74.1	58.8	72.1	57.3	69.4	6.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.8	70.8	61.6	53.9	65.9	60.9	51.9	61.2	59.7	27.3	74.5	26.3	72.3	25.3	69.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.4	14.1	12.1	-47.9	86.7	3.8	3.9	-50.6	89.5	-52.9	91.8	-55.0	94.0	-57.7	96.8
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	-6.0	0.9	14.5	33.1	46.5	56.1	59.2	54.7	44.4	30.2	7.1	-4.0
	DBStd	26.48	19.27	17.23	15.36	9.01	6.78	5.51	4.34	5.61	6.49	9.30	15.33	17.36
	HDD50	8656	1735	1374	1101	506	151	8	1	18	187	616	1286	1673
	HDD65	13442	2200	1794	1566	956	575	274	186	320	617	1080	1736	2138
	CDD50	702	0	0	0	0	42	190	285	164	20	1	0	0
	CDD65	13	0	0	0	0	0	6	5	2	0	0	0	0
	CDH74	713	0	0	0	0	47	255	246	163	2	0	0	0
	CDH80	146	0	0	0	0	3	74	37	32	0	0	0	0
	WSAvg	4.0	2.2	2.6	4.3	5.0	5.6	5.2	5.0	4.6	4.7	3.9	2.8	1.9
	PrecAvg	9.9	0.5	0.3	0.2	0.2	0.9	1.4	2.3	1.6	1.2	0.7	0.4	0.4
(15) Precipitation	PrecMax	12.7	1.1	0.8	0.6	0.4	1.8	3.1	4.2	2.9	2.6	1.6	1.2	0.8
	PrecMin	6.8	0.0	0.0	0.0	0.0	0.0	0.3	1.0	0.3	0.0	0.1	0.0	0.1
	PrecStd	1.8	0.3	0.2	0.2	0.1	0.5	0.7	0.9	0.7	0.6	0.3	0.2	0.2
	PrecStd	1.8	0.3	0.2	0.2	0.1	0.5	0.7	0.9	0.7	0.6	0.3	0.2	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	40.9	42.1	52.8	66.3	78.7	87.8	84.3	84.3	70.2	58.7	43.8	39.9	
	0.4% MCWB	37.7	35.4	40.8	53.0	53.0	59.8	60.0	59.4	53.0	47.6	37.5	34.7	
	2% DB	35.1	37.2	45.3	58.0	74.4	80.7	79.5	78.7	65.0	52.0	37.1	34.8	
	2% MCWB	31.6	32.7	35.5	42.5	51.6	56.0	58.4	57.9	51.5	43.6	33.3	31.5	
	5% DB	30.2	32.4	41.3	52.9	69.3	75.8	76.5	73.8	61.8	47.9	31.7	28.7	
	5% MCWB	28.1	28.4	33.1	39.3	49.2	54.3	56.8	56.4	49.4	40.3	30.0	27.1	
	10% DB	22.0	26.9	37.4	49.5	64.8	71.9	73.4	69.2	58.4	43.8	26.8	20.6	
	10% MCWB	21.2	24.4	30.7	37.7	47.6	53.7	56.1	54.6	47.9	38.7	25.8	19.8	
	0.4% WB	38.1	36.6	43.2	46.3	55.2	60.5	61.5	61.3	55.9	46.2	35.8	35.3	
	0.4% MCDB	41.9	42.0	56.0	57.8	76.9	80.1	73.8	77.1	67.4	53.4	39.5	38.9	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.1	32.6	39.4	42.1	53.2	57.7	60.5	59.6	53.1	43.1	32.5	32.0	
	2% MCDB	35.1	37.7	49.6	53.5	73.1	74.5	72.8	75.8	64.6	48.4	35.4	35.6	
	5% WB	29.9	27.7	36.0	40.4	51.5	55.9	59.5	57.9	50.8	41.1	29.4	27.7	
	5% MCDB	32.5	30.7	44.3	51.4	69.6	71.0	71.5	72.1	61.5	46.4	31.1	29.8	
	10% WB	25.5	22.6	33.1	38.8	49.5	54.4	58.1	56.2	49.1	38.7	25.8	20.7	
	10% MCDB	27.0	24.7	40.9	49.3	65.3	68.5	69.2	68.3	58.3	43.0	27.0	21.4	
(35) Mean Daily Temperature Range	MDBR	17.0	24.7	29.5	26.7	28.5	27.1	25.1	24.6	23.5	17.3	15.2	15.9	
	5% DB	24.2	25.8	28.0	31.4	38.7	35.4	33.0	34.6	29.0	22.9	19.9	23.8	
	5% WB	21.5	19.5	18.5	16.9	18.9	14.3	12.3	14.2	16.0	14.8	17.5	17.7	
	MCDBR	24.3	24.4	26.4	26.7	37.5	28.5	24.8	29.0	26.8	19.5	20.8	20.0	
	MCWBR	21.3	19.5	15.7	15.7	18.1	12.5	10.8	13.6	15.2	13.4	18.0	17.7	
	MCWBR	21.3	19.5	15.7	15.7	18.1	12.5	10.8	13.6	15.2	13.4	18.0	17.7	
(40) Clear-Sky Solar Irradiance	taub	0.224	0.249	0.260	0.343	0.367	0.372	0.372	0.367	0.303	0.273	0.205	0.173	
	taud	2.005	2.247	2.288	2.126	2.327	2.352	2.371	2.399	2.560	2.481	2.070	1.932	
	Ebn at Noon	142	233	275	265	270	270	266	258	261	227	154	106	
	Edh at Noon	12	19	26	39	35	35	33	30	21	16	10	6	
(44) All-Sky Solar Radiation	RadAvg	96	360	919	1531	1739	1798	1610	1282	817	373	123	46	
	RadStd	6	22	47	75	81	106	90	100	74	42	12	3	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	N/A
Regional (5 neighbors)	N/A	+3.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page